Robin Hur

(484) 995-5440 | robinhur2006@gmail.com | Glen Mills, PA 19342 | linkedin.com/in/robinkhur

Education

University of Virginia, School of Engineering and Applied Science **Charlottesville, VA | 2024–2027**
B.S in Mechanical Engineering, GPA: 3.72

- **Relevant coursework:** Statics, Strength of Materials, Thermodynamics, Fluid Mechanics, Dynamics, Material Science, Computational Methods, Mechatronics

Garnet Valley High School **Glen Mills, PA | 2020-2024**
Class of 2024, GPA: 4.31/4.50

Skills

- **Design & Analysis:** SOLIDWORKS (CSWA), Autodesk Fusion, GD&T, Python, MATLAB, Bambu Studio, Microsoft Office, FEA, Fabrication & Rapid Prototyping, Technical Documentation
- **Languages:** English (native), Korean (fluent), French (intermediate)

Experience

Undergraduate Research Assistant, *UVA Dept. of Material Science* **Feb 2026 – Present**

- Contributing to research on environmentally assisted cracking in aluminum alloys for aerospace applications with relevance to NASA and F-35 aircraft programs
- Working towards developing a Python app that integrates Alicat MC 10SLPM D/5M and Monasarch6184 010 Cal THProbe to precisely control nitrogen and temperature for environmentally assisted cracking experiments.

Undergraduate Research Assistant, *UVA Dept. of Chemical Engineering* **Aug 2024 – Nov 2025**

- Conducted all-atom molecular dynamics simulations using AMBER to study hydrogel formation at the atomic level
- Developed and managed 100+ simulation setup files with systematic organization workflows
- Automated Python analysis workflows to generate standardized CSV outputs with Git-based version control while generating and visualizing molecular trajectories in VMD to study structural evolution and interactions
- Produced publication-quality figures and supplementary data; contributed writing, data interpretation, and curated datasets to a manuscript under review

Research Outputs

- Publication: https://doi.org/10.26434/chemrxiv-2026-8jxvq (ChemRxiv preprint) **Jan 2026**

Undergraduate Teacher Assistant, *UVA Dept. of Electrical and Computer Engineering* **Jan 2026 – Present**

- Support students during and outside of the classroom by answering questions and reinforcing course material
- Assist with grading assignments, providing timely and constructive feedback
- Work closely with course instructors and Graduate head TAs to help prepare and manage course materials

Bridge Program Counselor, *UVA School of Engineering and Applied Sciences* **Jul – Aug 2025**

- Mentored 38 first-year engineering students as part of a 6-member counselor team during a 3.5-week residential program, guiding academic, social, and personal development
- Served as a Resident Advisor, cultivating a supportive community, promoting student engagement
- Tutored pre-calculus to enhance foundational skills and prepare students for college

Projects

Mechatronics and Robotics Society, *Mechanical Subteam* **Aug 2025 – Present**

- Designing and building a robot for the NASA Lunabotics competition
- Developed CAD models for a Lunabotics-style robot capable of manipulating lunar regolith
- Researched vibration frequency effects in granular material flow to optimize excavation subsystem performance

Potential Energy Propulsion Racer, *Project Member* **Nov 2025**

- Collaborated on an engineering team to design and build a 3D-printed, potential-energy-powered racing vehicle
- Applied CAD modeling, mechanical transmission design, and prototyping to develop and optimize vehicle components
- Delivered final technical report and curated comprehensive inventory of CAD parts, assemblies, and design files